Sanctions Risk and Dollar Convenience

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Abstract

Financial sanctions exploit the dollar network and alter reserve managers' demand for that network. I model this feedback. Accessibility risk subtracts from dollar convenience. Network complementarity amplifies reserve reallocation, while heterogeneous adjustment costs generate a matrix transition system. A policymaker who chooses sanctions after reserve managers fix portfolios ignores the future network loss. A public reputation state supports a restrained rule when discounted privilege exceeds the gain from deviation. I calibrate local slopes from reserve shares and convenience yields, then report global thresholds across nine function classes that share the same local equilibrium. The baseline break-even breadth lies between 5.2 and 5.8 percent across those closures. Public TIC holdings, aggregate COFER, and convenience yields identify direct responses and bound the network multiplier. The policy rule determines the future capacity of the instrument.

JEL codes: F31, F33, F51, E42, G15.
Keywords: financial sanctions, reserve currencies, convenience yield, network externalities, time inconsistency.

1 Introduction

In the last days of February 2022 the United States and its allies froze roughly $300 billion of assets belonging to the central bank of Russia: about half of that country's reserves, and the first such action ever taken against a central bank of the Group of Twenty (Ribakova et al., 2025). The freeze worked through accounts and custodians. It worked because Russia, like nearly every country, held its safe assets inside the currencies and custodial systems of a small group of Western jurisdictions. The power of the instrument comes from the size of the dollar network. The question of this paper is what exercising that power costs its owner.

The cost, if there is one, operates through a belief. A reserve manager outside the United States alliance network watched February 2022 and learned that the safety of dollar assets is state contingent: safe in most states of the world, inaccessible in the state where his government collides with Washington. Treasury yields and default risk stayed fixed. What changed was the probability each holder attaches to being able to use them when they are most needed. That probability times the expected haircut is a wedge, denominated in the same units as the convenience yield that makes reserve assets worth holding at a premium in the first place (Krishnamurthy and Vissing-Jorgensen, 2012). This paper prices the wedge, traces it through the network externality that sustains a dominant currency, and asks how a hegemon that understands the whole chain should use, or decline to use, the financial weapon.

The local benchmark has a sharp decomposition. Countries choose the dollar share of reserves by balancing convenience against portfolio concentration. Expected inaccessibility subtracts from convenience in the same units. With common portfolio curvature and network exposure, the aligned-exposed difference recovers the direct wedge and aggregate demand responds by $m = 1/(1 - \ell_1/\kappa)$. Heterogeneity turns this scalar into two distributional objects, A_z for direct exposure and A_λ for network loading. Their ratio links country responses to the aggregate network effect.

Adjustment costs turn a permanent reassessment into slow portfolio drift. The heterogeneous Euler system yields a stable transition matrix. Aggregate persistence exceeds cross-sectional persistence exactly when the dominant stable mode loads more heavily on total dollar demand than on exposure contrasts. The homogeneous two-type model gives this ranking analytically and supplies the baseline timing.

The hegemon's problem contains a commitment force. After reserve managers fix portfolios, the crisis payoff favors maximal intensity. Before they choose portfolios, the policy rule internalizes the privilege lost through future reserve diversification. The two-state reputation model supports the restrained rule when the discounted flow gap exceeds the current deviation gain. Forgiveness shortens punishment and raises the required patience.

The quantitative section reports a sanctions-use frontier. The issuer captures a share ζ_H of reserve convenience, and crisis liquidity scales the value of inaccessible reserves by χ_i. At breadth 0.02, full intensity requires a geopolitical benefit of 71.4 billion dollars per crisis. At breadth 0.10, the threshold is 363.9 billion. Nine matched global closures place the corresponding break-even breadth between 5.2 and 5.8 percent.

Bianchi and Sosa-Padilla (2025) establish that anticipated financial sanctions reduce dollar convenience and dollar holdings. I take that anticipation response as the direct demand block and close the policy loop through an endogenous currency network. Current use changes future demand, future demand changes the value of restraint, and restraint preserves future coercive capacity. The network block builds on Gopinath and Stein (2021), Farhi and Maggiori (2018), and He et al. (2019). Public TIC holdings discipline country responses. Aggregate COFER and convenience yields bound amplification.

The linear economy governs small reassessments around the observed reserve equilibrium. Bounded global closures govern routine use. Geopolitical benefits enter through threshold functions, which keeps the policy comparison in observable dollar units.

2 Reserves, Convenience, and the Sanctions Wedge

The linear-quadratic structure is the local economy that delivers the results. Propositions 1 through 5 are exact inside that economy. The sign results use the slope of convenience in network size. The magnitudes use the linear form and the calibration in Section 5. Appendix C extends the same logic to corners.

2.1 Environment

There is a continuum of countries $i \in [0,1]$ and a dominant-currency issuer, the hegemon. Each period, country i allocates its reserves between dollar assets, in share $a_i \in [0,1]$, and an outside reserve asset: gold, or any other store of value beyond the hegemon's legal reach. Dollar assets pay a financial return spread $\bar{r}$ over the outside asset (zero in the baseline calibration) and deliver liquidity services

$$\ell(N) = \ell_0 + \ell_1 N, \qquad \ell_1 \geq 0, \qquad N = \int_0^1 a_i \, di, \tag{1}$$

where N is the aggregate dollar share: the size of the dollar network. The dependence of ℓ on N is the reduced form of the complementarities that the dominant-currency literature microfounds, deeper markets and invoicing that matches the reserve asset (Gopinath and Stein, 2021; Farhi and Maggiori, 2018; He et al., 2019). Linearity gives exact local results and makes measurement target a slope. The local multiplier is scalar. Section 5 introduces global curvature and enumerates its fixed points. Around the observed equilibrium, the linear economy is the first-order approximation to every smooth closure considered below.

Sanctions enter as an expected loss of liquidity services. Country i has exposure $z_i \in [0,1]$. Exposed countries hold a mass μ of reserves, while countries inside the hegemon's security orbit hold the rest. With probability ω each period a geopolitical crisis arrives. The hegemon's rule assigns intensity $\bar{s} \in [0,1]$. Conditional on a crisis, the hegemon targets country i with probability p_i and blocks a fraction ϕ_i of its dollar reserves. The loss is most costly when reserve demand peaks. Let

$$\hat{\pi}_i \;=\; \omega \, \bar{s} \, p_i \, \phi_i \, \chi_i, \qquad \chi_i \equiv \frac{u_i'(L_i^{\text{crisis}})}{u_i'(L_i^{\text{normal}})} \geq 1, \tag{2}$$

where χ_i is the crisis-liquidity multiplier. The object $z_i\hat{\pi}_i$ is the *sanctions wedge*. It is a state-contingent haircut on the liquidity value of dollar reserves, measured in annual return units. This is the anticipation channel in Bianchi and Sosa-Padilla (2025). Reserve demand prices the product in (2). Quantity responses, crisis exposures, and liquidity-state evidence jointly identify its components.

Country i's per-period objective over its reserve share is quadratic:

$$u_i(a, a_{-1}) \;=\; a\big(\bar{r} + \lambda_i \ell(N) - z_i \hat{\pi}_i\big) \;-\; \frac{\kappa_i}{2} a^2 \;-\; \frac{\nu_i}{2}(a - a_{-1})^2. \tag{3}$$

The parameter λ_i measures exposure to the dollar network. The parameter κ_i prices concentration costs arising from currency mismatch, trade invoicing, and political diversification. The parameter

ν_i prices gradual portfolio adjustment. The homogeneous benchmark sets $\lambda_i = 1$, $\kappa_i = \kappa$, $\nu_i = \nu$, and $\hat{\pi}_i = \hat{\pi}$ for exposed countries. Set $\nu = 0$ for now. Section 3 restores it.

2.2 Pricing the weapon

In the homogeneous benchmark, the interior first-order condition gives

$$a_i = \frac{\bar{r} + \ell(N) - z_i \hat{\pi}}{\kappa}. \tag{4}$$

Proposition 1 (The sanctions wedge is priced like a tax on convenience)**.** *At any interior equilibrium: (i) the dollar share of an exposed country falls one-for-one with $\hat{\pi}$ scaled by $1/\kappa$, holding N fixed. (ii) The equilibrium difference between aligned and exposed portfolios is*

$$a_A - a_E = \frac{\hat{\pi}}{\kappa},$$

independent of the network term $\ell(N)$. (iii) The effective convenience yield earned by an exposed country is $\ell(N) - \hat{\pi}$: sanctions risk and liquidity services are the same object with opposite signs.

The literature measures convenience yields in basis points (Krishnamurthy and Vissing-Jorgensen, 2012; Du et al., 2018). Equation (2) places accessibility risk in the same units. With $\chi = 1$, a freeze that arrives once every twenty years, strikes a given country with probability 0.02 conditional on a crisis, and blocks one half of its portfolio creates a five-basis-point wedge. A crisis multiplier of two doubles that wedge without changing the actuarial probability.

2.3 The network multiplier

Aggregating (4) over countries and solving for N:

$$N = \frac{\bar{r} + \ell_0 - \mu\hat{\pi}}{\kappa - \ell_1}, \qquad \text{so} \qquad \frac{dN}{d\hat{\pi}} = -\frac{\mu}{\kappa} m, \quad m \equiv \frac{1}{1 - \ell_1/\kappa}. \tag{5}$$

Proposition 2 (Multiplier and uniqueness)**.** *If $\ell_1 < \kappa$ and the solution is interior, the equilibrium exists and is unique. A sanctions-risk shock that lowers exposed demand directly by $\mu\hat{\pi}/\kappa$ lowers the aggregate network by m times that amount, where $m > 1$ whenever $\ell_1 > 0$. If $\ell_1 \geq \kappa$, the contraction argument fails. Interior equilibria can be unstable and the constrained economy can display corner equilibria or multiplicity.*

The quantitative local economy satisfies $\ell_1 < \kappa$. Appendix C treats constrained choices. The inequality supplies a sufficient uniqueness condition. Root enumeration determines the equilibrium count outside it.

Corollary 1 (Micro and macro)**.** *In the homogeneous benchmark the cross-sectional difference $a_A - a_E = \hat{\pi}/\kappa$ differences out the common network term. The aggregate response in (5) is m times the direct response. Identification of m requires the cross-sectional response together with an aggregate quantity or convenience-yield moment.*

The homogeneous cancellation identifies the controls required in country comparisons. Heterogeneous network loadings and portfolio curvature enter explicitly below.

2.4 Heterogeneous reserve managers

Set $\nu_i = 0$ and retain interior choices. The policy is

$$a_i = \frac{\bar{r} + \lambda_i \ell(N) - z_i \hat{\pi}_i}{\kappa_i}. \tag{6}$$

Define $A_\lambda = \int_0^1 \lambda_i/\kappa_i \, di$ and, for a common proportional change in sanction intensity, $A_z = \int_0^1 z_i \omega p_i \phi_i \chi_i/\kappa_i \, di$.

Proposition 3 (Heterogeneous local response)**.** *If $1 - \ell_1 A_\lambda > 0$, the local equilibrium is unique and*

$$\frac{dN}{d\bar{s}} = -\frac{A_z}{1 - \ell_1 A_\lambda}. \tag{7}$$

The direct semi-elasticity of country i is $-z_i \omega p_i \phi_i \chi_i/\kappa_i$. A group comparison also contains differences in λ_i/κ_i multiplied by the equilibrium network response.

Equation (7) is the empirical object. Countries with the same geopolitical exposure can respond differently because crisis liquidity, portfolio curvature, and reliance on dollar invoicing differ. Aligned countries respond through $\lambda_i \ell_1 dN$. Their movement identifies the network slope jointly with the distribution of λ_i/κ_i and the common-shock process.

The dynamic model supplies the pace, and the pace carries the identification. That is the next section.

3 Dynamics: Reassessment and Slow Erosion

Restore heterogeneous adjustment costs $\nu_i > 0$ and let countries share discount factor β_c. The Euler equation is

$$\bar{r} + \lambda_i \ell(N_t) - z_i \hat{\pi}_{i,t} - \kappa_i a_{i,t} - \nu_i (a_{i,t} - a_{i,t-1}) + \beta_c \nu_i (\mathbb{E}_t a_{i,t+1} - a_{i,t}) = 0. \tag{8}$$

For a finite approximation with reserve-weight vector w, collect shares in $\mathbf{a}_t$, network loadings in $\boldsymbol{\lambda}$, and define $D_\nu = \mathrm{diag}(\nu_i)$ and $D_\kappa = \mathrm{diag}(\kappa_i)$. Deviations from a stationary equilibrium obey

$$B\tilde{\mathbf{a}}_{t+1} - C\tilde{\mathbf{a}}_t + D_\nu \tilde{\mathbf{a}}_{t-1} = 0, \qquad B = \beta_c D_\nu, \qquad C = D_\kappa + (1+\beta_c) D_\nu - \ell_1 \boldsymbol{\lambda} w'. \tag{9}$$

Proposition 4 (Heterogeneous transition system)**.** *Let*

$$\mathcal{M} = \begin{bmatrix} B^{-1}C & -B^{-1}D_\nu \\ I & 0 \end{bmatrix}.$$

Suppose $\mathcal{M}$ has exactly n eigenvalues inside the unit circle and the top block X of their eigenvectors is nonsingular. The unique bounded perfect-foresight path satisfies

$$\widetilde{\mathbf{a}}_{t+1} = A_s \widetilde{\mathbf{a}}_t, \qquad A_s = X \Lambda_s X^{-1}.$$

For any observable $r' \mathbf{a}_t$, its asymptotic persistence is

$$\rho(r) = \max\{|\lambda_j| : r' x_j \neq 0\}.$$

Aggregate dollar demand is more persistent than a cross-sectional exposure contrast $g' \mathbf{a}_t$ exactly when $\rho(w) > \rho(g)$.

The proposition unifies the heterogeneous static response and dynamic adjustment. The rank-one network term in C moves modes that load on aggregate reserve demand. Heterogeneous curvature and adjustment costs determine how strongly each stable mode loads on the aggregate and on a given country contrast.

Corollary 2 (Homogeneous two-type saddle path)**.** *Suppose $\ell_1 < \kappa$. After a permanent change in $\hat{\pi}$, the unique bounded perfect-foresight path satisfies*

$$N_t - N_\infty = \lambda_N \, (N_{t-1} - N_\infty), \qquad d_t - d_\infty = \lambda_d \, (d_{t-1} - d_\infty),$$

where $\lambda_N \in (0,1)$ is the stable root of $\beta_c \nu \lambda^2 - (\kappa + (1+\beta_c)\nu - \ell_1)\lambda + \nu = 0$ and λ_d solves the same equation with $\ell_1 = 0$. The aggregate is more persistent than the cross-section, $\lambda_N > \lambda_d$, and λ_N is increasing in ℓ_1: complementarity adds inertia as well as amplification.

A permanent reassessment moves reserve shares toward a new plateau. A transitory risk shock produces reversion. Country holdings estimate the contrast modes, while valuation-adjusted COFER estimates the aggregate mode (Arslanalp et al., 2022). The homogeneous restriction $\lambda_N > \lambda_d$ becomes the observable-mode restriction $\rho(w) > \rho(g)$ in the heterogeneous system.

Who, then, should fire such a weapon, and how often? The demand side is now complete, and the question moves to the hegemon.

4 The Hegemon and the Weaponization Bias

4.1 The hegemon's objective

Foreign official investors accept a return lower by $\ell(N)$ than their outside opportunity. Only a share of that saving reaches the issuer. Intermediation rents, regulatory costs, and foreign holdings outside direct sovereign liabilities absorb the remainder. Let $\zeta_H \in [0,1]$ denote the issuer's capture rate. With W denoting world reserves, the privilege flow is

$$\Omega(N) \;=\; \zeta_H \ell(N) \, N \, W, \tag{10}$$

which is increasing in N in the maintained local region. Geopolitical crises arrive with probability ω per period. In a crisis the hegemon chooses sanction intensity $s \in [0, 1]$ against the target and collects

$$G(s) \;=\; \gamma\left(s - \tfrac{1}{2}s^2\right), \tag{11}$$

where γ is a policy input. The functional form normalizes the ex post optimum to $s = 1$ when reserve managers hold portfolio beliefs fixed.

The timing within a period is the natural one and does all the work. Reserve managers set portfolios first on the basis of the anticipated rule $\bar{s}$ through the wedge (2). The crisis arrives after the portfolio choice. Today's sanction moves tomorrow's portfolios through what it teaches about the rule.

4.2 Discretion and the rule

Consider the benchmark in which the hegemon chooses s crisis by crisis and a single action cannot revise reserve managers' beliefs.

Assumption 1 (Fixed beliefs at the crisis date)**.** *At the date of a crisis, portfolios and the anticipated future rule $\bar{s}$ are predetermined. The current choice of s does not move either object within the period.*

Under Assumption 1, the ex post problem is $\max_s G(s)$ and its solution is $s_D = 1$. This allocation supplies the punishment state in the reputation model. When public actions revise beliefs, the continuation value enters the crisis decision.

The Ramsey hegemon chooses the rule itself, internalizing $N(\bar{s})$ from (5):

$$\max_{\bar{s}\in[0,1]} \; V(\bar{s}) \;=\; \Omega\big(N(\bar{s})\big) + \omega\, G(\bar{s}). \tag{12}$$

Proposition 5 (Weaponization bias)**.** *Let $c \equiv \mu p\phi\chi/(\kappa - \ell_1)$ and $\bar{N}_0 = (\bar{r} + \ell_0)/(\kappa - \ell_1)$. Suppose the allocation is interior and $\gamma > 2\zeta_H \ell_1 \omega c^2 W$. The Ramsey rule is*

$$s_R \;=\; \frac{\gamma - \zeta_H c W(\ell_0 + 2\ell_1 \bar{N}_0)}{\gamma - 2\zeta_H \ell_1 \omega c^2 W} \tag{13}$$

clamped to $[0, 1]$. If $c > 0$ and the marginal captured privilege is positive at $N(1)$, then $s_R < s_D = 1$. The gap is the cost of taking portfolio beliefs as fixed at the crisis date.

The rule becomes more restrained when a given action destroys more captured privilege. This occurs when target breadth, freeze severity, crisis liquidity, issuer pass-through, or the network response rises. As $\gamma \to \infty$, the Ramsey and fixed-belief choices converge.

4.3 A solved reputation state

Public reputation takes two values. State R records adherence to the Ramsey rule and state P records punishment. Reserve managers price $\bar{s} = s_R$ in state R and $\bar{s} = 1$ in state P. Adherence keeps state R. A crisis deviation to $s = 1$ moves the next period to P. Punishment uses $s = 1$ and returns to R with forgiveness probability $\alpha \in [0, 1]$. Let $f_R = V(s_R)$ and $f_P = V(1)$ denote annual flow values from equation (12). The public values solve

$$U_R = f_R + \beta_H U_R, \tag{14}$$

$$U_P = f_P + \beta_H \big[(1-\alpha)U_P + \alpha U_R\big]. \tag{15}$$

Proposition 6 (Reputation with forgiveness)**.** *The two-state public strategy sustains s_R if and only if*

$$\frac{\beta_H}{1-\beta_H(1-\alpha)}(f_R - f_P) \geq G(1) - G(s_R). \tag{16}$$

Equivalently, the minimum discount factor is

$$\beta_H^*(\alpha) = \frac{G(1) - G(s_R)}{f_R - f_P + (1-\alpha)[G(1) - G(s_R)]}.$$

It rises with α. With permanent punishment, $\alpha = 0$, the interior local economy reduces to

$$\frac{\beta_H}{1-\beta_H} \geq \frac{\gamma}{\omega\gamma - 2\zeta_H \ell_1 \omega^2 c^2 W}.$$

At the baseline inputs, permanent punishment requires an annual discount rate no greater than 5.0 percent. Positive forgiveness raises the required patience because it shortens the loss of captured privilege after deviation.

5 Quantitative Analysis

5.1 Calibration

Table 1 separates observations and external estimates from model-implied objects and scenario inputs. This distinction governs the interpretation of every number below.

The observed pair $\ell(N_0) = 73$ basis points and $N_0 = 0.59$ implies model curvature $\kappa = 124$ basis points under the normalization $\bar{r} = 0$. The baseline wedge is 5.0 basis points when $p = 0.02$ and $\chi = 1$. Geopolitical quantities enter as arguments of the reported threshold functions.

5.2 Local and global economies

The linear model governs the 2022 reassessment and the proofs in Sections 2–4. Routine use calls for a global closure. The baseline closure replaces linear convenience with

$$\ell^g(N) = \ell_{\min} + \frac{\ell_{\max} - \ell_{\min}}{1 + \exp\{-b(N - \bar{N})\}} \tag{17}$$

Table 1: Calibration (annual).

Parameter		Value	Category	Basis
Convenience yield, pre-2022	$\ell(N_0)$	73 bp	external estimate	AAA–Treasury spread (KVJ)
Dollar share, pre-2022	N_0	0.59	observed aggregate	COFER
World reserves	W	\$12 tn	observed aggregate	COFER
Freeze severity	ϕ	0.5 $\{0.25, 0.75\}$	observed episode	Russia 2022 asset freeze
Diversification cost	κ	124 bp	model implied	$\kappa = \ell(N_0)/N_0$
Network multiplier	m	1.5 $\{1, 2.5\}$	scenario	identified by joint moments
Adjustment cost	ν	0.133	scenario implied	3-year aggregate half-life
Country discount factor	β_c	0.96	external convention	annual value
Exposed reserve share	μ	0.5 $\{0.3, 0.7\}$	scenario	coalition classification
Crisis arrival	ω	0.05	scenario	one event per two decades
Target breadth	p	0.02	scenario	threshold input
Crisis liquidity	χ	1 $\{1, 2, 4\}$	scenario	one is actuarial lower bound
Issuer capture	ζ_H	0.50 $\{0.25, 1\}$	scenario	privilege pass-through
Geopolitical stakes	γ	\$200 bn $\{50–800\}$	policy input	threshold reported below

Notes: KVJ is Krishnamurthy and Vissing-Jorgensen (2012). COFER is public in aggregate. Braces give the ranges for reference inputs.

and replaces quadratic portfolio cost by

$$C'(a) = qa + \tau a^3. \tag{18}$$

The code sets $\ell_{\min} = 0.001$, $\ell_{\max} = 0.012$, and $\tau = 0.002$. The remaining coefficients solve

$$\ell^g(N_0) = \ell(N_0), \qquad \ell^{g\prime}(N_0) = \ell_1, \qquad C''(N_0) = \kappa, \tag{19}$$

with $\bar{r} + \ell^g(N_0) - C'(N_0) = 0$. The comparison adds

$$\ell^H(N) = \ell_{\min} + (\ell_{\max} - \ell_{\min})\frac{N^\eta}{K^\eta + N^\eta}, \qquad \ell^P(N) = \ell_{\min} + AN^\alpha,$$

and linear, cubic, or quintic marginal portfolio costs. Every pairing matches $\ell(N_0)$, $\ell'(N_0)$, $C'(N_0)$, and $C''(N_0)$. The nine economies therefore share the observed equilibrium and the complete local response. Their differences measure global closure risk.

Proposition 7 (Global scenario equilibrium)**.** *Let ℓ^c be a differentiable global convenience closure and suppose $C_i''(a) > 0$ on $[0,1]$. If the slope of the aggregate best response satisfies*

$$\sup_{N\in[0,1]} \left\{ \ell^{c\prime}(N) \int_{\mathcal{I}(N)} \frac{\lambda_i}{C_i''(a_i(N))}\, di \right\} < 1, \tag{20}$$

where $\mathcal{I}(N)$ is the set of interior reserve managers, the constrained global equilibrium exists and is unique. The logistic baseline satisfies this condition over the full unit interval. The numerical routine enumerates all fixed points for every alternative closure.

The global routine-use results are nonlinear scenarios. The local 2022 numbers remain exact comparative statics within the linear economy.

Table 2: Break-even breadth across matched global closures

Convenience function	Quadratic cost	Cubic marginal cost	Quintic marginal cost
Logistic	5.8%	5.5%	5.5%
Bounded Hill	5.5%	5.2%	5.2%
Concave power	5.7%	5.4%	5.4%

Notes: each closure matches the same observed dollar share, convenience level, convenience slope, and local portfolio curvature. Entries give the breadth at which full intensity and restraint have equal flow value at the baseline policy inputs.

Table 3: Steady states: the price of the weapon by regime of use.

	Pre-2022 ($\hat{\pi} = 0$)	Single use ($p = 0.02$)	Soft restrictions ($\phi = 0.15$)	Routine use ($p = 25\%$)
Dollar share N (percent)	59	56.0	58.1	23.2
exposed a_E	59	53.9	57.5	0.0
aligned a_A	59	58.0	58.7	46.5
Convenience yield (bp)	73	71.7	72.6	58
Captured privilege ($bn/yr)	25.8	24.1	25.3	8.1

Notes: the single-use and soft-restriction columns use the local economy. The routine-use column uses the matched global economy. Captured privilege sets $\zeta_H = 0.50$. All geopolitical parameters are scenario inputs.

The break-even breadth lies between 5.2 and 5.8 percent across all nine closures. The routine-use dollar share ranges from 16.4 to 23.5 percent. Thus the policy crossing is stable while the distant quantity counterfactual remains sensitive to global curvature. The root-enumeration audit finds at most 1 equilibrium on the unit interval for these nine calibrations at routine-use breadth.

5.3 One freeze

Table 3 reports steady states, and Figure 1 the transition, for the single-use regime: the world learns in 2022 that the rule is $\bar{s} = 1$ with breadth $p = 0.02$.

In the single-use phase experiment, the aggregate dollar share declines 3.0 percentage points and captured privilege falls by $1.7 billion a year. At $m = 2.5$ the same wedge lowers the dollar share by 5.1 points. At $m = 1$ it lowers the share by 2.0 points. Figure 2 shows the aligned-country response implied by network feedback. That response is zero when $\ell_1 = 0$.

5.4 The sanctions-use frontier

Figure 3 reports the sanctions-use frontier from the global economy. At the baseline inputs, full intensity and full restraint have the same flow value at $p^* = 5.5$ percent. This crossing maps each belief about target breadth into a policy ranking. At $p = 25$ percent, the global equilibrium has $N = 23.2$ percent and captured privilege is lower by $17.8 billion a year.

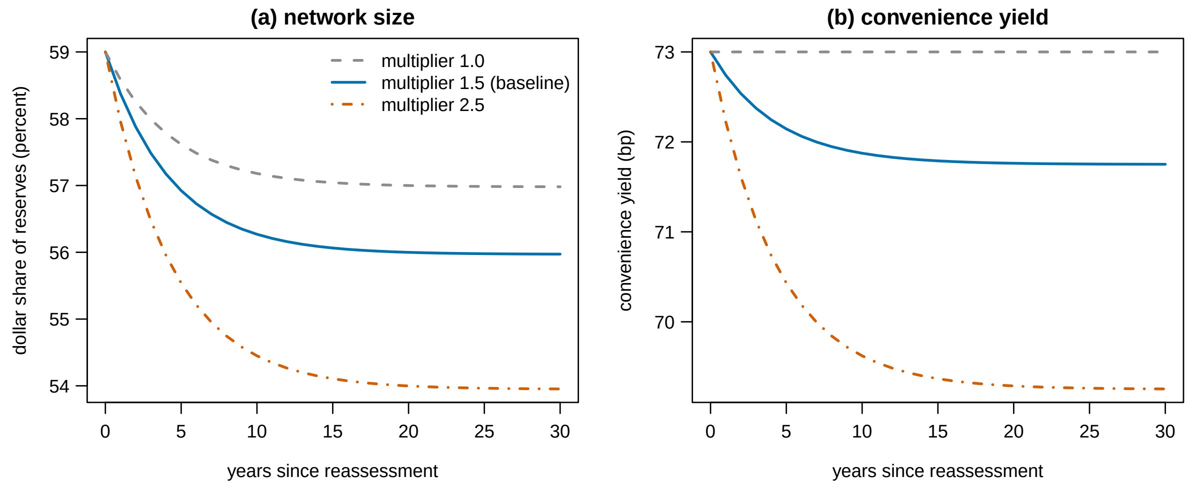

Figure 1: Transition after the 2022 reassessment (single-use regime, $\hat{\pi}$: $0 \to 5.0$ bp). Aggregate dollar share and convenience yield, for network multipliers 1.0, 1.5, 2.5.

The threshold is the function $p^*(\gamma, \phi, \mu, m, \zeta_H, \chi)$. Holding other inputs at baseline, it falls from 10.7 to 2.7 percent as m rises from 1 to 2.5. It falls from 11.1 to 3.7 percent as ϕ rises from 0.25 to 0.75. Increasing χ from one to two lowers it from 5.5 to 2.8 percent. At $\chi = 4$ it is 1.4 percent. Changing ζ_H from 0.25 to one moves the crossing from 11.0 to 2.8 percent. With $\gamma = 800$ billion, full intensity remains above restraint throughout $p \in [0, 1]$, so no crossing exists on the feasible interval. The full reported scenario grid gives thresholds from 0.3 to 34.4 percent where a crossing exists.

The inverse threshold $\gamma^*(p, \phi, \mu, m, \zeta_H, \chi)$ is the geopolitical value required to justify full intensity. At baseline inputs it equals \$71.4 billion for $p = 0.02$, \$363.9 billion for $p = 0.10$, and \$710.0 billion for $p = 0.25$. This representation leaves γ as a policy input and states the model's tradeoff without assigning it an empirical value.

5.5 Rules against discretion

Figure 4 evaluates the local rule at alternative values of the policy input γ. At \$200 billion, $s_R = 0.83$ and the fixed-belief benchmark is one. The flow value of commitment is \$0.1 billion a year. At $\gamma = 50$ billion the rule is 0.30. At $\gamma = 800$ billion it is 0.96. The figure uses the local approximation and $\zeta_H = 0.50$.

The quantitative exercise establishes thresholds conditional on reserve demand, network curvature, crisis liquidity, and issuer pass-through. Estimating those objects is therefore more important than attaching confidence to a single baseline number.

6 Empirical Designs

The heterogeneous model turns on the distribution of direct responses and the network feedback in (7). No single regression identifies both. The empirical program therefore combines country

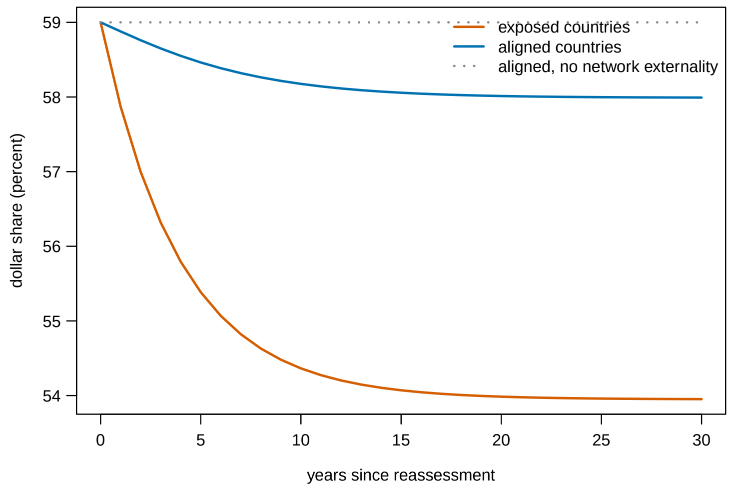

Figure 2: The network fingerprint: exposed and aligned dollar shares after the reassessment, baseline multiplier. The dotted path is the aligned share when $\ell_1 = 0$: with the network externality shut down, aligned countries stay fixed.

holdings, aggregate reserve composition, and convenience-yield responses while treating custodial attribution as a measurement problem.

Design 1: country reserve demand. The main public outcome is foreign official holdings of United States Treasury securities reported by TIC, scaled by total reserves from the IMF International Financial Statistics. The event-study equation is

$$\Delta\left(\frac{\mathrm{UST}_{i,t}}{\mathrm{Reserves}_{i,t}}\right) = \alpha_i + \tau_t + \sum_q \sum_{k\neq -1} \beta_{qk} Z_{iq} \mathbf{1}\{t = t_0 + k\} + \Gamma' X_{i,t} + \varepsilon_{i,t}. \tag{21}$$

The vector Z_{iq} enters in separate dimensions. It contains pre-2022 political alignment, prior financial-sanctions exposure, reserve adequacy, dollar debt, and dollar invoicing. Reporting the full coefficient path for each dimension preserves their distinct economic content and removes arbitrary index weights. Directly sanctioned targets form a separate treatment group because their holdings reflect legal restrictions as well as portfolio choice. Country and time effects accompany commodity terms, reserve accumulation, exchange-rate valuation, sovereign risk, and global safe-asset demand.

TIC reports the location of the foreign holder or custodian, not beneficial ownership in every case (United States Department of the Treasury, 2026). Custody centers are pooled into a separate group and omitted in a robustness sample. Official and private holdings are separated when the published table permits. The disclosed central-bank subsample and gold purchases provide validation outcomes. COFER is not used at the country level because the IMF keeps individual submissions confidential (International Monetary Fund, 2026).

Design 2: aggregate network moments. Let Sal_t measure public announcements of asset freezes and payment restrictions in the Global Sanctions Data Base release 4 (Global Sanctions

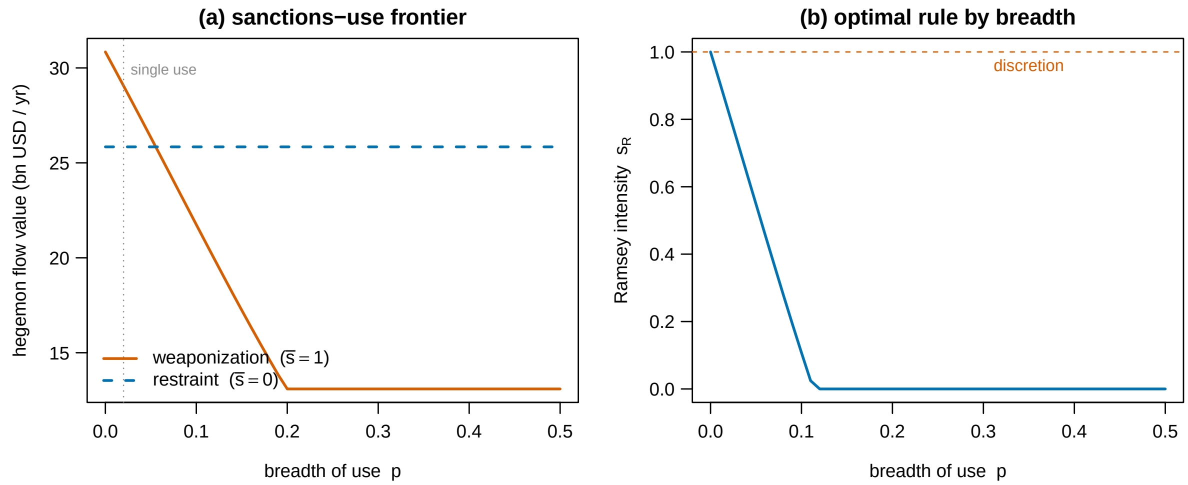

Figure 3: The sanctions-use frontier in the matched global economy. Panel (a) compares full intensity with restraint as anticipated breadth varies. Panel (b) reports the optimal commitment intensity.

Data Base, 2025). The price equation is

$$CY_t \;=\; \rho\, CY_{t-1} + \delta\, \mathrm{Sal}_t + \Xi' Z_t + u_t \tag{22}$$

where Z_t includes global risk, monetary policy, Treasury supply, dealer balance sheets, and exchange-rate hedging costs. The sign restriction is $\delta < 0$. Multiple financial-sanctions episodes supply time variation, while trade-only sanctions serve as a placebo category. Valuation-adjusted COFER measures the aggregate quantity response. Dollar convenience relative to other safe currencies measures the price response.

Let $[D_L, D_U]$ denote the confidence set for the positive magnitude of the direct country response and $[T_L, T_U]$ the set for the aggregate decline. The local model maps them into

$$\mathcal{M} = \left[\max\left\{1, \frac{T_L}{D_U}\right\}, \frac{T_U}{D_L}\right]. \tag{23}$$

Aligned-country responses and convenience yields tighten this set by restricting A_λ. The empirical result is the joint identified set for amplification.

Structural mapping. Country responses discipline A_z and the distribution of direct semi-elasticities. Aligned responses and aggregate quantities discipline A_λ. The stable roots of equation (9) discipline the distribution of adjustment costs. Treasury interest savings relative to the global convenience flow bound ζ_H. Reserve-adequacy stress episodes bound χ by comparing the marginal value of liquid reserves in crisis and normal periods. Joint uncertainty in these objects maps directly into a set for the policy threshold. Gold purchases and disclosed central-bank portfolios supply overidentifying moments.

Table 4 records the restrictions. Trade-only sanctions isolate general geopolitical news from asset inaccessibility. Aligned holdings measure network transmission. The joint system converts

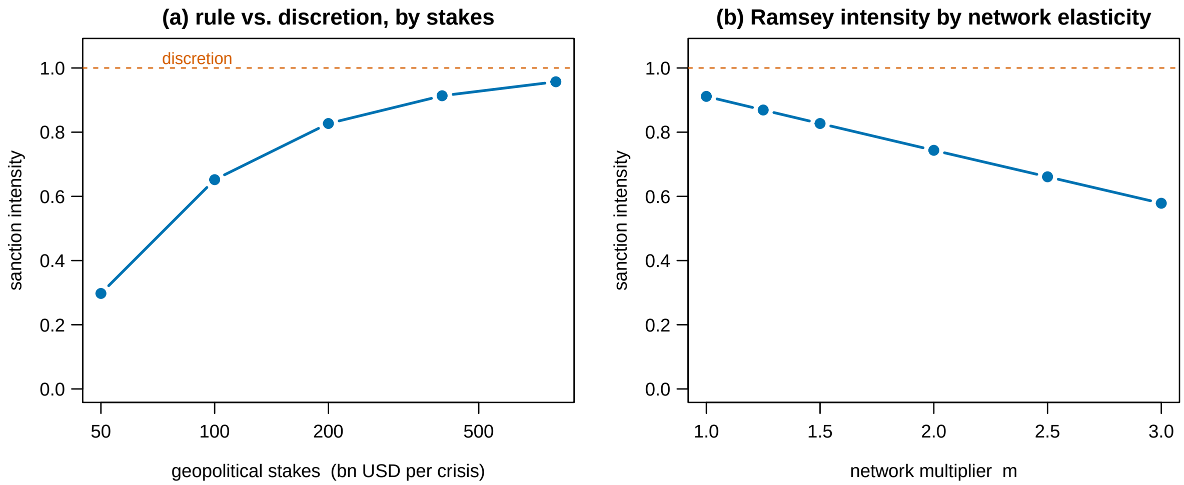

Figure 4: The weaponization bias. Panel (a): Ramsey intensity by geopolitical stakes γ (log scale). Discretion is 1 throughout. Panel (b): Ramsey intensity by network multiplier at $\gamma = 200$.

these responses into the set in equation (23).

Gold purchases and aggregate COFER provide validation moments (World Gold Council, 2024; Arslanalp et al., 2022). Table 5 records the public-data implementation.

7 Conclusion

Financial sanctions draw strength from the same reserve network that they can weaken. The paper closes that policy loop. In the local economy, accessibility risk enters reserve demand as a haircut on convenience and network complementarity amplifies the aggregate response. With heterogeneous countries, raw group differences no longer reveal a single portfolio curvature. The relevant object is the weighted response in (7).

The heterogeneous transition matrix connects that local response to reserve dynamics. Country contrasts identify direct adjustment modes. Aggregate COFER identifies the network mode. Their joint response bounds the multiplier and determines how quickly current sanctions reduce future dollar demand.

The fixed-belief crisis allocation chooses maximal intensity because current portfolios are predetermined. The Ramsey rule values the future network. A two-state public reputation sustains that rule when the discounted privilege gap covers the current deviation gain. Forgiveness raises the patience threshold.

Nine global closures place the baseline break-even breadth between 5.2 and 5.8 percent. Crisis liquidity and issuer capture move the wider threshold set. TIC holdings, aggregate COFER, gold purchases, and dollar convenience yields discipline those objects. The resulting policy frontier measures how present use changes the future value of the financial weapon.

Table 4: Predictions and implementation.

Prediction	Model magnitude	Null	Estimation
Exposed reserve demand falls after accessibility reassessment	$-z_i \omega p_i \phi_i \chi_i / \kappa_i$ locally	0	TIC and disclosures
Response rises with predetermined exposure	dose response	flat	event study
Asset freezes exceed trade-only sanctions	larger absolute response	equal	GSDB split
Dollar convenience falls with financial-sanctions salience	negative cumulative response	0	yield series
Aligned holdings move with the network	sign of $\ell_1 \Delta N$	0 if $\ell_1 = 0$	reserve panel
Permanent and transitory components differ	persistent component	full reversion	event path

Notes: predictions are local sign and mapping restrictions. TIC measures custodial location imperfectly. COFER supplies only aggregate currency composition.

Table 5: Data and implementation.

Dataset	Coverage	Frequency	Access	Observed or proxied	Measure
IMF COFER	aggregate shares, 1999–	quarterly	public	observed (aggregate only)	in hand
TIC official holdings	by country, monthly, custodial bias	monthly	public	proxy for beneficial ownership	in hand
Disclosed CB composition	subsample of central banks	annual	public reports	observed, selected sample	assembled from reports
WGC gold purchases	by country	quarterly	public	observed	in hand
GSDB release 4	1950–2023	episode	public academic	observed	in hand
UN ideal points	all members, 1946–	annual	public	observed	in hand
Convenience-yield series	US, G10	monthly	replicable	observed	in hand

Notes: the salience index uses public GSDB events. Disclosed reserve composition is selected because publication is voluntary. TIC reports custodial country attribution. Beneficial ownership enters through disclosed central-bank portfolios.

A Proofs

A.1 Proof of Proposition 1

In the homogeneous benchmark, $\kappa > 0$ makes (3) strictly concave in a. The interior first-order condition gives (4). Holding N fixed gives $\partial a_i/\partial\hat{\pi} = -z_i/\kappa$. Subtracting the exposed policy from the aligned policy removes the common network term and gives $a_A - a_E = \hat{\pi}/\kappa$. The exposed first-order condition can be written $\kappa a_E = \bar{r} + [\ell(N) - \hat{\pi}]$, which proves the convenience-yield statement. ■

A.2 Proof of Proposition 2

Define the aggregate best response $B(N) = [\bar{r} + \ell_0 + \ell_1 N - \mu\hat{\pi}]/\kappa$. On the interior region its Lipschitz constant is ℓ_1/κ. If $\ell_1 < \kappa$, the contraction theorem gives a unique fixed point. Solving it gives (5). Total differentiation yields $dN/d\hat{\pi} = -(\mu/\kappa)/(1 - \ell_1/\kappa)$. When $\ell_1 \geq \kappa$, this proof supplies no uniqueness conclusion. The constrained fixed-point problem remains continuous and therefore has an equilibrium, but its number and location depend on intercepts and binding portfolio bounds. ■

A.3 Proof of Corollary 1

The cross-sectional equality follows from Proposition 1. Proposition 2 gives the aggregate response. Their ratio is m in the homogeneous interior economy. Recovering that ratio requires both moments. ■

A.4 Proof of Proposition 3

Strict concavity gives the unique interior policy (6). Differentiating it with respect to $\bar{s}$ and integrating gives

$$\frac{dN}{d\bar{s}} = \ell_1 A_\lambda \frac{dN}{d\bar{s}} - A_z.$$

The maintained inequality makes the denominator positive and gives (7). For country i, the derivative holding N fixed is $-z_i\omega p_i\phi_i\chi_i/\kappa_i$. The total derivative adds $(\lambda_i\ell_1/\kappa_i)dN/d\bar{s}$. Hence a group contrast removes the network term only when the relevant distributions of λ_i/κ_i coincide. ■

A.5 Proof of Proposition 4

Stack equation (9) with the identity that carries $\widetilde{\mathbf{a}}_t$ into the lag block. This gives

$$\begin{bmatrix} \widetilde{\mathbf{a}}_{t+1} \\ \widetilde{\mathbf{a}}_t \end{bmatrix} = \mathcal{M} \begin{bmatrix} \widetilde{\mathbf{a}}_t \\ \widetilde{\mathbf{a}}_{t-1} \end{bmatrix}.$$

Exactly n stable roots and the transversality condition select the stable invariant subspace. Let its eigenvector matrix be $[X'\ Y']'$ and its eigenvalue matrix be Λ_s. Nonsingularity of X lets current shares index the stable coordinates. Hence

$$\widetilde{\mathbf{a}}_{t+1} = X\Lambda_s X^{-1}\widetilde{\mathbf{a}}_t.$$

Writing x_j for column j of X, the observable $r'\widetilde{\mathbf{a}}_t$ contains precisely the modes for which $r'x_j \neq 0$. Its asymptotic decay rate is the largest modulus among those roots. Applying this definition to w and g proves the last statement. ■

A.6 Proof of Corollary 2

Averaging (8) over countries and collecting terms gives

$$\beta_c\nu\, N_{t+1} - (\kappa + (1+\beta_c)\nu - \ell_1)N_t + \nu N_{t-1} = -(\bar{r} + \ell_0 - \mu\hat{\pi}),$$

and subtracting the aligned from the exposed equation gives the same recursion for d_t with $\ell_1 = 0$ and forcing $\hat{\pi}$ (the network term is common and cancels). Let $P(\lambda) = \beta_c\nu\lambda^2 - B\lambda + \nu$ with $B = \kappa + (1+\beta_c)\nu - \ell_1$. Then $P(0) = \nu > 0$ and $P(1) = \beta_c\nu - B + \nu = \ell_1 - \kappa < 0$, so one real root lies in $(0,1)$. The product of the roots is $1/\beta_c > 1$, and their sum is $B/(\beta_c\nu) > 0$, so the second root is positive and exceeds one. There is exactly one stable root $\lambda \in (0,1)$, and the unique bounded solution given the predetermined initial condition is the stated first-order recursion, with the constant equal to the new steady state. The stable root

$$\lambda = \frac{B - \sqrt{B^2 - 4\beta_c\nu^2}}{2\beta_c\nu}$$

is decreasing in B. The implicit derivative gives the same sign: at the stable root the polynomial crosses from positive to negative, so $P_\lambda(\lambda) < 0$ and $d\lambda/dB = \lambda/P_\lambda < 0$. Since B is decreasing in ℓ_1, $\lambda_N > \lambda_d$ and $\partial\lambda_N/\partial\ell_1 > 0$. ■

A.7 Proof of Proposition 5

Under fixed beliefs, $G'(s) = \gamma(1-s)$ and $s_D = 1$. For commitment, substitute $N(\bar{s}) = \bar{N}_0 - \omega c\bar{s}$, where $c = \mu p\phi\chi/(\kappa - \ell_1)$, into (12). The second derivative is $2\zeta_H\ell_1\omega^2c^2W - \omega\gamma$, which is negative under the stated condition. The first-order condition uses $\Omega'(N) = \zeta_H W(\ell_0 + 2\ell_1 N)$ and solves to (13). Rearrangement gives

$$1 - s_R = \frac{\zeta_H cW[\ell_0 + 2\ell_1 N(1)]}{\gamma - 2\zeta_H\ell_1\omega c^2 W} > 0.$$

The inequality uses the positive marginal captured privilege at $N(1)$. The value of commitment is the quadratic vertex gap

$$V(s_R) - V(1) = \tfrac{1}{2}(\omega\gamma - 2\zeta_H\ell_1\omega^2c^2W)(1-s_R)^2 > 0. \qquad ■$$

A.8 Proof of Proposition 6

Solving the two Bellman equations in (15) gives

$$U_R - U_P = \frac{f_R - f_P}{1 - \beta_H(1-\alpha)}.$$

Adherence at a crisis keeps state R. Deviation gains $G(1) - G(s_R)$ immediately and enters state P next period. The one-shot deviation principle therefore gives

$$G(1) - G(s_R) \leq \beta_H(U_R - U_P),$$

which is equation (16). Solving the inequality for β_H gives $\beta_H^*(\alpha)$. Its denominator falls with α, so the threshold rises. At $\alpha = 0$, the quadratic vertex formula from Proposition 5 gives

$$f_R - f_P = frac12(\omega\gamma - 2\zeta_H \ell_1 \omega^2 c^2 W)(1 - s_R)^2.$$

Since $G(1) - G(s_R) = \gamma(1 - s_R)^2/2$, canceling the common factor yields the displayed permanent-punishment condition. ∎

A.9 Proof of Proposition 7

For any N, strict convexity of C_i gives a unique constrained reserve choice. Projection onto $[0, 1]$ is continuous, so the aggregate best response maps the unit interval into itself and has a fixed point. At an interior choice, its derivative with respect to N is $\lambda_i \ell^{c\prime}(N)/C_i''(a_i)$. It is zero at a strict corner. Integrating gives the expression inside braces in (20). The strict upper bound makes the aggregate map a contraction and proves uniqueness. ∎

B The Saddle Path and Adjustment-Cost Calibration

The characteristic quadratic of Proposition 2 can be inverted for the adjustment cost that delivers a target persistence: setting $P(\lambda) = 0$ and solving for ν,

$$\nu \;=\; \frac{(\kappa - \ell_1)\,\lambda}{(1-\lambda)(1-\beta_c\lambda)},$$

so a three-year aggregate half-life ($\lambda = 2^{-1/3}$) pins $\nu = 0.133$ at the baseline (κ, ℓ_1). The pair (N_t, d_t) recovers the full transition in levels through $a_{A,t} = N_t - \mu d_t$ and $a_{E,t} = a_{A,t} + d_t$. Different roots drive the two recursions. The exposed-aligned gap, with root λ_d and half-life 2.4 years, closes on its plateau faster than the aggregate share, whose root λ_N has half-life 3 years. Design 1 and the COFER aggregate can check this joint timing prediction.

C Corners and Tipping

With common κ and $\ell_1 < \kappa$, projection onto $[0, 1]$ weakly lowers the slope of the local aggregate best response. The constrained local equilibrium is therefore unique. The routine-use calculation uses (17) and (18), with Proposition 7 supplying its uniqueness condition. Failure of the contraction inequality admits multiple fixed points. The estimated slope of the aggregate best response then determines whether tipping enters the policy set.

D Parameter Discipline and Sensitivity

Table 6 records the status and range of each object. At $\ell_1 = 0$, aligned portfolios do not respond through the network. When $\omega p\phi\chi$ approaches zero, the accessibility wedge vanishes. As γ grows, the commitment and fixed-belief allocations converge.

Table 6: Parameter discipline and sensitivity.

Parameter	Enters at	Units	Source or moment	Range examined	Effect of the range
$\ell(N_0) = 73$ bp	eq. (1)	bp per year	external estimate	observed uncertainty	privilege scales with yield
$N_0 = 0.59$	eq. (5)	share	aggregate COFER	observed uncertainty	pins κ with $\ell(N_0)$
$\kappa = 124$ bp	eq. (4)	bp per share	implied, $\kappa = \ell(N_0)/N_0$	-30% (outside-asset case)	erosion 3.0 $\rightarrow$ 5.5 pp
$m\ (\ell_1) = 1.5$	eq. (5)	unitless	scenario, joint moments	1 to 2.5	erosion 2.0–5.1 pp
$\nu = 0.133$	eq. (8)	utility	scenario half-life	timing alternatives	steady states unaffected
$\beta_c = 0.96$	eq. (8)	annual	externally fixed	annual discount	negligible near 1
$\mu = 0.5$	eq. (5)	reserve share	scenario	0.3 to 0.7	threshold 9.1–4.0%
$\omega = 0.05$	eq. (2)	per year	scenario	alternative arrival rates	wedge and G scale
$\phi = 0.5$	eq. (2)	share frozen	observed episode	0.25 to 0.75	threshold 11.1–3.7%
$p = 0.02$	eq. (2)	probability	scenario	0 to 1	sanctions-use frontier
$\chi = 1$	eq. (2)	ratio	scenario lower bound	1 to 4	threshold 5.5–1.4%
$\zeta_H = 0.50$	eq. (10)	share	scenario	0.25 to 1	threshold 11.0–2.8%
$W = \$12$ tn	eq. (10)	USD	aggregate COFER	observed uncertainty	dollar magnitudes scale
$\gamma = \$200$ bn	eq. (11)	USD per crisis	policy input	50 to 800	s_R 0.30–0.96

Notes: only the product $\omega\bar{s}p\phi\chi$ enters homogeneous reserve demand. Policy thresholds depend separately on the arrival rate because geopolitical benefits are realized in crisis states.

E Data Appendix

Reserve composition. IMF COFER reports quarterly aggregate currency shares of allocated reserves (International Monetary Fund, 2026). Individual-country reports are confidential. I valuation-adjust the aggregate dollar series before using it as a quantity moment (Arslanalp et al., 2022).

Country-level holdings. TIC reports monthly foreign official holdings by reported country (United States Department of the Treasury, 2026). This is custodial attribution and need not equal beneficial ownership. Custody hubs are aggregated and then omitted in a robustness sample. Country reserve totals come from IMF International Financial Statistics. Gold purchases come from World Gold Council releases (World Gold Council, 2024).

Sanctions. Global Sanctions Data Base release 4 covers episodes through 2023 (Global Sanctions Data Base, 2025). The salience index retains asset freezes and payment restrictions. Trade-only measures form a placebo group. The 2022 freeze magnitude is documented separately (Ribakova et al., 2025).

Alignment. Ideal-point distances from Bailey et al. (2017), updated through the sample. The exposure measure is the pre-2022 average distance from the G7 mean, so that exposure is predetermined at the event.

Convenience yields. The Krishnamurthy and Vissing-Jorgensen (2012) AAA–Treasury spread, the Du et al. (2018) Treasury premium (five-year, G10 average and by currency), and the Jiang et al. (2021) Treasury basis. Global controls: VIX, term premia, and Federal Reserve balance-sheet variables.

F Extensions

Endogenous outside asset. Let the outside asset's issuer invest to raise its own network quality, lowering κ over time. The outside-asset counterfactual of Section 5 is the comparative static: each step down in κ raises the erosion per use and lowers the Ramsey intensity. A full dynamic treatment makes κ_t a state in the hegemon's problem and turns the model into a duopoly of networks, the race sketched by Farhi and Maggiori (2018).

The target's margin. Freezing a sovereign's reserves degrades its crisis insurance and can push it toward default, imposing credit losses on the hegemon's own financial system. Bianchi and Sosa-Padilla (2024) model the margin. In the present notation those losses subtract from G, reducing γ and, by Proposition 5, widening the gap between discretion and any rule a planner would choose.

Continuous exposure. Proposition 3 already permits $z_i \in [0, 1]$. Its direct response is proportional to $z_i \omega p_i \phi_i \chi_i / \kappa_i$. A monotone raw dose response requires conditional control of the remaining components.

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